



Experiment No.: 6

Title: Use string functions and regex for email validation, data extraction, and text analysis.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code, Google Chrome

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>String Functions & Email Analyzer</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>String Functions & Email Analyzer</h1>
    <p class="subtitle">
      Analyze text using JavaScript String Functions and Regular Expressions
    </p>
    <textarea
      id="textInput"
      placeholder="Enter your text here..."
    >
      Example:
      Hello Shruti!
      Contact me at shruti@gmail.com
      You can also email test@example.com
      Invalid emails: abc@gmail, hello@, @domain.com">
      Hello Shruti!
      Contact me at shruti@gmail.com
      You can also email test@example.com
      Student email: student@college.edu
      Invalid emails:
      abc@gmail
      hello@
      @domain.com
    </textarea>
    <div class="search-box">
      <input
        type="text"
        id="searchInput"
        placeholder="Enter a word/text to search using includes()"
      >
    </div>
  </div>
```



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```
</div>
<button onclick="analyzeText()">
  Analyze Text
</button>
<div class="results">
  <div class="card">
    <h3>1. Uppercase</h3>
    <p id="uppercase" class="output"></p>
  </div>
  <div class="card">
    <h3>2. Lowercase</h3>
    <p id="lowercase" class="output"></p>
  </div>
  <div class="card">
    <h3>3. String Length</h3>
    <p id="length" class="output"></p>
  </div>
  <div class="card">
    <h3>4. includes() Search</h3>
    <p id="includes" class="output"></p>
  </div>
  <div class="card">
    <h3>5. Email Validation using Regex</h3>
    <div id="emailSummary" class="output"></div>
    <h4>Valid Emails</h4>
    <div id="validEmails"></div>
    <h4>Invalid Emails</h4>
    <div id="invalidEmails"></div>
  </div>
  <div class="card">
    <h3>6. Text Analysis using split()</h3>
    <p id="wordCount" class="output"></p>
    <h4>Words</h4>
    <div id="words"></div>
  </div>
</div>
</div>
</div>
<script src="script.js"></script>
</body>
</html>
```

Script.js:

```
function analyzeText() {
  const text =
    document.getElementById("textInput").value;
```



```
const searchText =
  document.getElementById("searchInput").value;
document.getElementById("uppercase").textContent =
  text.toUpperCase();
document.getElementById("lowercase").textContent =
  text.toLowerCase();
document.getElementById("length").textContent =
  "Number of characters: " + text.length;
if (searchText.trim() === "") {
  document.getElementById("includes").textContent =
    "Please enter something to search.";
} else {
  const found =
    text.toLowerCase()
      .includes(searchText.toLowerCase());
  if (found) {
    document.getElementById("includes").innerHTML =
      `<span class="valid">
        "${searchText}" was found in the text.
      </span>`;
  } else {
    document.getElementById("includes").innerHTML =
      `<span class="invalid">
        "${searchText}" was not found in the text.
      </span>`;
  }
}
const emailRegex =
  /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
const tokens =
  text.trim() === ""
    ? []
    : text.trim().split(/\s+/);
let validEmails = [];
let invalidEmails = [];
tokens.forEach(function(token) {
  const email =
    token.replace(/[,!?:;()+$%/, "");
  if (email.includes("@")) {
    if (emailRegex.test(email)) {
      validEmails.push(email);
    } else {
      invalidEmails.push(email);
    }
  }
})
```



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```
}
});
const totalEmails =
    validEmails.length + invalidEmails.length;
document.getElementById("emailSummary").innerHTML = `
    <div class="count-box">
        <div class="count">
            <strong>Total Emails:</strong>
            ${totalEmails}
        </div>
        <div class="count valid">
            Valid Emails:
            ${validEmails.length}
        </div>
        <div class="count invalid">
            Invalid Emails:
            ${invalidEmails.length}
        </div>
    </div>
`;
const validContainer =
    document.getElementById("validEmails");
validContainer.innerHTML = "";
if (validEmails.length === 0) {
    validContainer.innerHTML =
        '<span class="invalid">No valid emails found.</span>';
} else {
    validEmails.forEach(function(email) {
        const span =
            document.createElement("span");
        span.className = "email";
        span.textContent = email;
        validContainer.appendChild(span);
    });
}
const invalidContainer =
    document.getElementById("invalidEmails");
invalidContainer.innerHTML = "";
if (invalidEmails.length === 0) {
    invalidContainer.innerHTML =
        '<span class="valid">No invalid emails found.</span>';
} else {
    invalidEmails.forEach(function(email) {
        const span =
```



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Experiment-6

```
        document.createElement("span");
        span.className = "invalid-email";
        span.textContent = email;
        invalidContainer.appendChild(span);
    });
}
const words =
    text.trim() === ""
        ? []
        : text.trim().split(/\s+/);
document.getElementById("wordCount").textContent =
    "Total words: " + words.length;
const wordContainer =
    document.getElementById("words");
wordContainer.innerHTML = "";
words.forEach(function(word) {
    const span =
        document.createElement("span");
        span.className = "word";
        span.textContent = word;
        wordContainer.appendChild(span);
});
}
```

Style.css:

```
* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
    font-family: Arial, sans-serif;
}
body {
    background: linear-gradient(135deg, #96d0c7, #6ebf94);
    min-height: 100vh;
    padding: 30px;
}
.container {
    max-width: 950px;
    margin: auto;
    background: white;
    padding: 30px;
    border-radius: 15px;
    box-shadow:
        0 10px 30px rgba(0, 0, 0, 0.2);
}
```



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```
}
h1 {
  text-align: center;
  color: #333;
  margin-bottom: 10px;
}
.subtitle {
  text-align: center;
  color: #666;
  margin-bottom: 25px;
}
textarea {
  width: 100%;
  height: 160px;
  padding: 15px;
  border: 2px solid #ddd;
  border-radius: 10px;
  resize: vertical;
  font-size: 16px;

  outline: none;
}
textarea:focus {
  border-color: #667eea;
}
.search-box {
  margin-top: 15px;
}
input {
  width: 100%;
  padding: 12px;
  border: 2px solid #ddd;
  border-radius: 8px;
  font-size: 15px;
  outline: none;
}
input:focus {
  border-color: #667eea;
}
button {
  display: block;
  margin: 20px auto;
  padding: 12px 35px;
  border: none;
```



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Experiment-6

```
border-radius: 8px;
background: #667eea;
color: white;
font-size: 16px;
cursor: pointer;
}
button:hover {
  background: #5568d8;
}
.results {
  margin-top: 20px;
}
.card {
  background: #f5f6fa;
  margin: 15px 0;
  padding: 18px;
  border-radius: 10px;
  border-left: 5px solid #667eea;
}
.card h3 {
  color: #333;
  margin-bottom: 10px;
}
.card h4 {
  margin-top: 15px;
  margin-bottom: 8px;
  color: #444;
}
.output {
  color: #555;
  line-height: 1.6;
  word-wrap: break-word;
}
.email {
  display: inline-block;
  background: #667eea;
  color: white;
  padding: 6px 10px;
  margin: 4px;
  border-radius: 5px;
}
.invalid-email {
  display: inline-block;
  background: #ffdddd;
```



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```
color: red;
padding: 6px 10px;
margin: 4px;
border-radius: 5px;
}
.valid {
color: green;
font-weight: bold;
}
.invalid {
color: red;
font-weight: bold;
}
}
.word {
display: inline-block;
background: #e8eaff;
color: #444;
padding: 5px 8px;
margin: 3px;
border-radius: 4px;
}
}
.count-box {
display: flex;
gap: 15px;
flex-wrap: wrap;
}
}
.count {
padding: 12px 20px;
background: white;
border-radius: 8px;
box-shadow:
0 2px 8px rgba(0, 0, 0, 0.1);
}
}
```




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Experiment-6

Output:

String Functions & Email Analyz

Ask Gemini

C:/Users/hp/Documents/24070521148_ShrutiBawankar_C1/JavaScript_Lab/Experiments/EXP6/index.html

GmailYouTubeCourse: LLM Engine...DO457 - pr01s02HACKER HOUSE GO...All Bookmarks

String Functions & Email Analyzer

Analyze text using JavaScript String Functions and Regular Expressions

Hello Shruti!
Contact me at shruti@gmail.com
You can also email test@example.com
Student email: student@college.edu
Invalid emails:
abc@gmail
hello@
@domain.com

hello

Analyze Text

1. Uppercase

HELLO SHRUTI! CONTACT ME AT SHRUTI@GMAIL.COM YOU CAN ALSO EMAIL TEST@EXAMPLE.COM
STUDENT EMAIL: STUDENT@COLLEGE.EDU INVALID EMAILS: ABC@GMAIL HELLO@ @DOMAIN.COM

2. Lowercase

hello shruti! contact me at shruti@gmail.com you can also email test@example.com student email:



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Experiment-6

The screenshot displays a web application titled "String Functions & Email Analy" in the browser's title bar. The address bar shows the file path: C:/Users/hp/Documents/24070521148_ShrutiBawankar_C1/JavaScript_Lab/Experiments/EXP6/index.html. The application interface includes a blue "Analyze Text" button at the top. Below it, five analysis sections are visible:

- 1. Uppercase**: HELLO SHRUTI! CONTACT ME AT SHRUTI@GMAIL.COM YOU CAN ALSO EMAIL TEST@EXAMPLE.COM STUDENT EMAIL: STUDENT@COLLEGE.EDU INVALID EMAILS: ABC@GMAIL HELLO@ @DOMAIN.COM
- 2. Lowercase**: hello shruti! contact me at shruti@gmail.com you can also email test@example.com student email: student@college.edu invalid emails: abc@gmail hello@ @domain.com
- 3. String Length**: Number of characters: 169
- 4. includes() Search**: "hello" was found in the text.
- 5. Email Validation using Regex**:
 - Total Emails: 6
 - Valid Emails: 3
 - Invalid Emails: 3
 - Valid Emails list: shruti@gmail.com, test@example.com, student@college.edu
 - Invalid Emails list: abc@gmail, hello@, @domain.com
- 6. Text Analysis using split()**: Total words: 19

The "Words" section for item 6 shows the following words: Hello, Shruti!, Contact, me, at, shruti@gmail.com, You, can, also, email, test@example.com, Student, email:, student@college.edu, Invalid, emails:, abc@gmail, hello@, @domain.com.

Result: The webpage successfully analyzes the entered text using string functions, split(), and Regex, and identifies valid and invalid email IDs.

Conclusion: JavaScript string functions and regular expressions provide an efficient way to perform text analysis and email validation.